



Departamento de Engenharia Informática e de Sistemas
Licenciatura em Engenharia Informática
Licenciatura em Engenharia Informática - PL
Ramo de Desenvolvimento de Aplicações

Human Computer Interaction

Interacção Pessoa Máquina

Assignment No. 3: Interaction Design

15 points out of 20

GOAL

The objective of this project is the specification and implementation of the graphical interface and respective interaction of an application/game. The application will be subject to the discretion of the student, but it should be centred around “**Audio Guides**”. It is also considered fundamental that the application developed should allow the inclusion of at least one type of user with special needs (in terms of vision). This aspect can be substantiated through a theoretical study and a practical implementation of a form of connection with hardware that allows an alternative or innovative form of communication/interaction.

Sugestion:

Blind and visually impaired visitors experience a lot of constraints when visiting museums exhibitions, especially due to the lack of cognitive, physical and sensorial access to exhibits or replicas, increased by the inaccessibility to use the digital media technologies designed to provide different experiences, among other constraints.

The exploration of **Trackable Interactive Multi-modal Manipulatives** (TIMM) and multimodal **Tangible User Environment** (TUE) are also a possibility. This system provides a multimodal tangible user environment (TUE), enabling people with visual impairments to create, modify and naturally interact with graphical representations on a multitouch surface. The system supports a novel notion of active position, proximity, stacking, and orientation tracking of manipulatives.

TASKS/ PROCEDURE

Preparation of a report proposing a design of an interactive system and its implementation, for example, using Scratch containing the following topics:

- **Users Analysis**, user profiles characterization using suitable methods for example personas.



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- **Task Analysis**, preceded whenever possible by questionnaires. Correct identification of the tasks that the system under study must support (system functionality) based on fundamental questions of task analysis.
- **Metaphors**, defining the set of metaphors underlying the proposed Conceptual Model.
- **Low or High fidelity prototype** (To illustrate the idea through a prototyping tool, for example, using Balsamiq, or Axure Justinmind).
- **Implementation** using Scratch (available in <https://scratch.mit.edu/>).
- **Report** on a binding form **with hardware** that allows a **new and alternative interaction mode**.

All decisions made on application design must be justified from the point of view of usability and/or desired accessibility, namely:

- The inspiration for the development of the interface;
- The adopted interaction styles and objects;
- The composition of the interface;
- The icons used;
- The colours adopted;
- Human factor aspects;
- Other aspects considered important.

DEADLINE

The work (a zip file with the Report and the Application) must be delivered through the Moodle platform until the **8 of June**.